

MLP - multi-layer perceptron

Input data: $x \rightarrow [N, \Delta]$ $\Delta = 28 \times 28 = 784$

$t \rightarrow [N]$ $e \in \{0, 1, 2, \dots, 9\}$

Batch size $B \Rightarrow x \rightarrow [B, \Delta]$

$t \rightarrow [B]$

Rețea neurală: L straturi

$x^{(e)}$: intrările stratului e

$y^{(e)}$: ieșirile stratului e

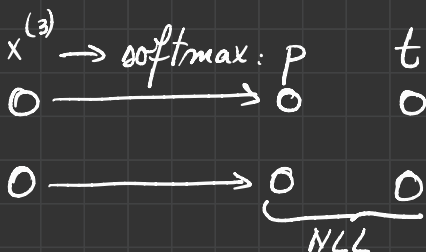
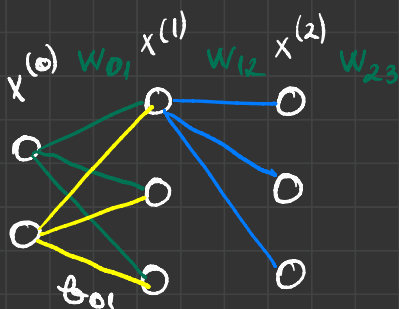
$\theta^{(e)}$: parametrii stratului e (weights + biases)

$\mathcal{L}$: funcție de cost (NLL)

2 etape: forward: $y^{(e)} = f_e(x^{(e)}, \theta^{(e)})$

backward

- primește: $\frac{\partial \mathcal{L}}{\partial y^{(e)}}$
- retine: $\frac{\partial \mathcal{L}}{\partial \theta^{(e)}} \Rightarrow \text{update } \theta^{(e)}$
- intoarce: $\frac{\partial \mathcal{L}}{\partial x^{(e)}}$



$$X^{(1)} = X^{(0)} \cdot W_{01} + b_{01} \quad (\text{linear layer})$$

$$X^{(0)} : [B, 2] \approx \mathbb{R}^{B \times M} \quad X^{(1)} : [B, 3] \approx \mathbb{R}^{B \times N} \Rightarrow W_{01} : [2, 3] \approx \mathbb{R}^{M \times N}; b : [3] \approx \mathbb{R}^M$$

$$X^{(2)} = X^{(1)} \cdot W_{12} + b_{12}$$

$$X^{(3)} = X^{(2)} \cdot W_{23} + b_{23}$$

linear forward : $x \cdot w + b$ → de la layer-ul anterior (deja calculat)

backward : $\frac{\partial L}{\partial x} = \left[\frac{\partial L}{\partial y} \right] \cdot \frac{\partial y}{\partial x}$

$$\frac{\partial L}{\partial w} = \left[\frac{\partial L}{\partial y} \right] \cdot \frac{\partial y}{\partial w}$$

$$\frac{\partial L}{\partial x} : \begin{matrix} B \times M & B \times D & D \times M \\ \uparrow & \uparrow & \uparrow \\ y = XW + b = \end{matrix} \begin{bmatrix} x_{11}, x_{12}, \dots, x_{1D} \\ x_{21}, x_{22}, \dots, x_{2D} \\ \vdots \\ x_{B1}, x_{B2}, \dots, x_{BD} \end{bmatrix} \begin{bmatrix} w_{11}, w_{12}, \dots, w_{1M} \\ w_{21}, w_{22}, \dots, w_{2M} \\ \vdots \\ w_{D1}, w_{D2}, \dots, w_{DM} \end{bmatrix} =$$

$$= \begin{bmatrix} x_{11} w_{11} + x_{12} w_{21} + \dots + x_{1D} w_{D1} + b_1 & x_{11} w_{1M} + x_{12} w_{2M} + \dots + x_{1D} w_{DM} + b_M \\ x_{21} w_{11} + x_{22} w_{21} + \dots + x_{2D} w_{D1} + b_1 & x_{21} w_{1M} + x_{22} w_{2M} + \dots + x_{2D} w_{DM} + b_M \\ \vdots & \vdots \\ x_{B1} w_{11} + x_{B2} w_{21} + \dots + x_{BD} w_{D1} + b_1 & x_{B1} w_{1M} + x_{B2} w_{2M} + \dots + x_{BD} w_{DM} + b_M \end{bmatrix}$$

$$\frac{\partial L}{\partial x_{11}} = \sum_{i=1}^B \sum_{j=1}^M \frac{\partial L}{\partial y_{i,j}} \cdot \frac{\partial y_{i,j}}{\partial x_{1,1}} = \left[\frac{\partial L}{\partial y} \right] \cdot \frac{\partial y}{\partial x_{1,1}}$$

→ de la layer anterior

$$\frac{\partial y}{\partial x_{11}} = \begin{bmatrix} w_{11} & w_{12} & \dots & w_{1M} \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \quad \frac{\partial y}{\partial x_{21}} = \begin{bmatrix} 0 & 0 & \dots & 0 \\ w_{11} & w_{12} & \dots & w_{1M} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

$$\frac{\partial L}{\partial y} = \begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \frac{\partial L}{\partial y_{12}} & \dots & \frac{\partial L}{\partial y_{1M}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L}{\partial y_{B1}} & \frac{\partial L}{\partial y_{B2}} & \dots & \frac{\partial L}{\partial y_{BM}} \end{bmatrix} \quad \begin{matrix} * \\ = \text{înmulțire element cu} \\ \text{element, urmata de sumă} = \end{matrix}$$

$$= \frac{\partial L}{\partial y_{11}} w_{11} + \frac{\partial L}{\partial y_{12}} w_{12} + \dots + \frac{\partial L}{\partial y_{1M}} w_{1M}$$

Analog pentru restul x_{ij}

$$\frac{\partial L}{\partial x} = \begin{bmatrix} \frac{\partial L}{\partial x_{11}} & \frac{\partial L}{\partial x_{12}} & \dots & \frac{\partial L}{\partial x_{1D}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L}{\partial x_{B1}} & \frac{\partial L}{\partial x_{B2}} & \dots & \frac{\partial L}{\partial x_{BD}} \end{bmatrix} =$$

$$\begin{bmatrix} \frac{\partial L}{\partial y_{11}} w_{11} + \dots + \frac{\partial L}{\partial y_{1M}} w_{1M} & \dots & \frac{\partial L}{\partial y_{11}} w_{D1} + \dots + \frac{\partial L}{\partial y_{1M}} w_{DM} \\ \vdots & \ddots & \vdots \\ \frac{\partial L}{\partial y_{B1}} w_{11} + \dots + \frac{\partial L}{\partial y_{BM}} w_{1M} & \dots & \frac{\partial L}{\partial y_{B1}} w_{D1} + \dots + \frac{\partial L}{\partial y_{BM}} w_{DM} \end{bmatrix} =$$

$$\begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \dots & \frac{\partial L}{\partial y_{1M}} \\ \vdots & & \vdots \\ \frac{\partial L}{\partial y_{B1}} & \dots & \frac{\partial L}{\partial y_{BM}} \end{bmatrix} \begin{bmatrix} w_{11} & w_{21} & \dots & w_{D1} \\ w_{12} & w_{22} & \dots & w_{D2} \\ \vdots & \vdots & & \vdots \\ w_{1M} & w_{2M} & \dots & w_{DM} \end{bmatrix} = \frac{\partial L}{\partial y} \cdot w^T$$

$$\frac{\partial L}{\partial w_{11}} \cdot \frac{\partial L}{\partial w_{11}} = \sum_{i=1}^B \sum_{j=1}^M \frac{\partial L}{\partial y_{ij}} \cdot \frac{\partial y_{ij}}{\partial w_{11}} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial w_{11}}$$

$$\frac{\partial y}{\partial w_{11}} = \begin{bmatrix} x_{11} & 0 & \dots & 0 \\ x_{21} & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ x_{B1} & 0 & \dots & 0 \end{bmatrix}^M \quad \frac{\partial L}{\partial w_{12}} = \begin{bmatrix} 0 & x_{11} & \dots & 0 \\ 0 & x_{21} & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & x_{B1} & \dots & 0 \end{bmatrix}^B$$

$$\frac{\partial L}{\partial y} = \begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \frac{\partial L}{\partial y_{12}} & \dots & \frac{\partial L}{\partial y_{1M}} \\ \vdots & \vdots & & \vdots \\ \frac{\partial L}{\partial y_{B1}} & \frac{\partial L}{\partial y_{B2}} & \dots & \frac{\partial L}{\partial y_{BM}} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial w_{11}} = \text{sumă de pe diagonala înmulțire element cu element} =$$

$$= \frac{\partial L}{\partial y_{11}} x_{11} + \frac{\partial L}{\partial y_{21}} x_{21} + \dots + \frac{\partial L}{\partial y_{B1}} x_{B1}$$

Analog pentru restul $w_{ij} \Rightarrow$

$$\frac{\partial L}{\partial \mathbf{w}} = \begin{bmatrix} \frac{\partial L}{\partial w_{11}} & \frac{\partial L}{\partial w_{12}} & \dots & \frac{\partial L}{\partial w_{1M}} \\ \frac{\partial L}{\partial w_{21}} & \frac{\partial L}{\partial w_{22}} & \dots & \frac{\partial L}{\partial w_{2M}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L}{\partial w_{D1}} & \frac{\partial L}{\partial w_{D2}} & \dots & \frac{\partial L}{\partial w_{DM}} \end{bmatrix} =$$

$$\frac{\partial L}{\partial y_{11}} x_{11} + \frac{\partial L}{\partial y_{21}} x_{21} + \dots + \frac{\partial L}{\partial y_{B1}} x_{B1}$$

$$\frac{\partial L}{\partial y_{1M}} x_{12} + \frac{\partial L}{\partial y_{2M}} x_{21} + \dots + \frac{\partial L}{\partial y_{BM}} x_{B1}$$

$$\frac{\partial L}{\partial y_{11}} x_{10} + \frac{\partial L}{\partial y_{21}} x_{20} + \dots + \frac{\partial L}{\partial y_{B1}} x_{B0}$$

$$\frac{\partial L}{\partial y_{1M}} x_{10} + \frac{\partial L}{\partial y_{2M}} x_{20} + \dots + \frac{\partial L}{\partial y_{BM}} x_{B0}$$

$[B \times M]$

$[D \times B]$

$$= \begin{bmatrix} x_{11} & x_{21} & \dots & x_{B1} \\ x_{12} & x_{22} & \dots & x_{B2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1D} & x_{2D} & \dots & x_{BD} \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \frac{\partial L}{\partial y_{12}} & \dots & \frac{\partial L}{\partial y_{1M}} \\ \frac{\partial L}{\partial y_{21}} & \frac{\partial L}{\partial y_{22}} & \dots & \frac{\partial L}{\partial y_{2M}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L}{\partial y_{B1}} & \frac{\partial L}{\partial y_{B2}} & \dots & \frac{\partial L}{\partial y_{BM}} \end{bmatrix}$$

$$= \mathbf{x}^T \frac{\partial L}{\partial \mathbf{y}}$$

$$\frac{\partial L}{\partial \theta} = \left[\frac{\partial L}{\partial y} \right] \cdot \frac{\partial y}{\partial \theta}$$

↳ stîut de la layerul anterior

$$\frac{\partial \mathcal{L}}{\partial \mathbf{B}_1} = \sum_{i=1}^B \sum_{j=1}^M \frac{\partial \mathcal{L}}{\partial y_{ij}} \cdot \frac{\partial y_{ij}}{\partial \mathbf{B}_1}$$

$$\frac{\partial y}{\partial \theta} = \underbrace{\begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 0 \end{bmatrix}}_M B$$

$$\frac{\partial L}{\partial y} = \begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \dots & \frac{\partial L}{\partial y_{1m}} \\ \vdots & & \vdots \\ \frac{\partial L}{\partial y_{B1}} & \dots & \frac{\partial L}{\partial y_{Bm}} \end{bmatrix}$$

$\rightarrow$ = înmulțire element cu element, urmată de sumă $\Rightarrow$

$$\Rightarrow \frac{\partial y}{\partial B_1} = \frac{\partial L}{\partial y_{11}} + \frac{\partial L}{\partial y_{21}} + \dots + \frac{\partial L}{\partial y_{B_1}}$$

Analog pentru restul $b_i \Rightarrow$

$$\Rightarrow \frac{\partial L}{\partial \mathbf{B}} = \begin{bmatrix} \frac{\partial L}{\partial B_1} \\ \vdots \\ \frac{\partial L}{\partial B_M} \end{bmatrix} = \begin{bmatrix} \frac{\partial L}{\partial y_{11}} + \frac{\partial L}{\partial y_{21}} + \dots + \frac{\partial L}{\partial y_{B_1}} \\ \vdots \\ \frac{\partial L}{\partial y_{1M}} + \frac{\partial L}{\partial y_{2M}} + \dots + \frac{\partial L}{\partial y_{B_M}} \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \frac{\partial L}{\partial y_{21}} & \dots & \frac{\partial L}{\partial y_{B1}} \\ \frac{\partial L}{\partial y_{12}} & \frac{\partial L}{\partial y_{22}} & \dots & \frac{\partial L}{\partial y_{B2}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L}{\partial y_{1M}} & \frac{\partial L}{\partial y_{2M}} & \dots & \frac{\partial L}{\partial y_{BM}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1_B \end{bmatrix}$$

$$= \frac{\partial L}{\partial y}^T \cdot \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1_B \end{bmatrix}$$

$\nwarrow$ $\text{np.ones}((B,))$

ReLU

$$y = \max(x, 0)$$

↓
rezultat de la
un layer linear



$$\frac{\partial L}{\partial x} = \left[\frac{\partial L}{\partial y} \right] \cdot \frac{\partial y}{\partial x} ; \quad \frac{\partial y}{\partial x} = \frac{\partial}{\partial x} \max(x, 0) =$$

↳ stint

$$= \frac{\partial}{\partial x} \begin{cases} 0, & x < 0 \\ x, & x > 0 \end{cases} = \begin{cases} 0, & x < 0 \\ 1, & x > 0 \end{cases} \Rightarrow$$

$$\Rightarrow \frac{\partial L}{\partial x} = \begin{cases} 0, & x < 0 \\ \frac{\partial L}{\partial y}, & x > 0 \end{cases}$$

Cross-entropy loss

predictie $y \rightarrow [B, M]$, nr. clase

$$\text{softmax: } p_k = \frac{e^{y_k}}{\sum_j e^{y_j}}$$

↳ probabilitatea clasei k pentru predictia y

$$\mathcal{L} = \frac{1}{B} \sum_{(x,t) \in \text{Batch}} -\log p_t(x, \theta) =$$

$$= \sum_j^m -y_j \log p_j \Rightarrow \mathcal{L}_i = -\log p_{i,t}$$

$\hookrightarrow 1$ if $j = t$ else 0

target

$$\text{numpy: } -\text{np.log}(p[\text{range}(\text{batch}), t])$$

$$\frac{\partial \mathcal{L}}{\partial y}$$

$$\mathcal{L} = -\log p_k$$

$$p_k = \text{softmax}(y_k) = \frac{e^{y_k}}{\sum_j e^{y_j}}$$

$$\frac{\partial \mathcal{L}}{\partial y_j}$$

$$= \frac{\partial \mathcal{L}}{\partial p_k}$$

$$\cdot \frac{\partial p_k}{\partial y_j}$$

$$\frac{\partial \mathcal{L}}{\partial p_k}$$

$$= -\frac{1}{p_k}$$

$$\frac{\partial p_k}{\partial y_j}$$

caz 1: $j \neq k \Rightarrow e^{y_k}$ constantă $\Rightarrow$ derivată calculată doar pentru $\frac{1}{\sum_j e^{y_j}}$

$$\frac{\partial p_k}{\partial y_j} = e^{y_k} \cdot \left(-\frac{1}{(\sum_{j'} e^{y_{j'}})^2} \right) \cdot e^{y_j}$$

caz 2: $j = k$

$$\frac{\partial p_k}{\partial y_j} = \frac{e^{y_k} \cdot \left(-\frac{1}{(\sum_{j'} e^{y_{j'}})^2} \right) - (e^{y_k})^2}{(\sum_{j'} e^{y_{j'}})^2}$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial y_j} = \frac{\partial \mathcal{L}}{\partial p_k} \cdot \frac{\partial p_k}{\partial y_j} =$$

caz 1:

$$-\frac{1}{p_k} \cdot e^{y_k} \cdot \frac{-1}{\left(\sum_{j'} e^{y_{j'}}\right)^2} \cdot e^{y_j} = \frac{\sum_{j'} e^{y_{j'}}}{e^{y_k}} \cdot \frac{e^{y_k} e^{y_j}}{\left(\sum_{j'} e^{y_{j'}}\right)^2}$$

Diagram showing the derivation of the first term:

$$\frac{p_k}{\downarrow e^{y_k}} \bigg| \frac{e^{y_k}}{\sum_{j'} e^{y_{j'}}$$

Diagram showing the softmax function:

$$\frac{e^{y_j}}{\sum_{j'} e^{y_{j'}}} \rightarrow \text{softmax}$$

caz 2:

$$-\frac{1}{p_k} \cdot e^{y_k} \cdot \frac{\sum_{j'} e^{y_{j'}} - (e^{y_k})^2}{\left(\sum_{j'} e^{y_{j'}}\right)^2} =$$

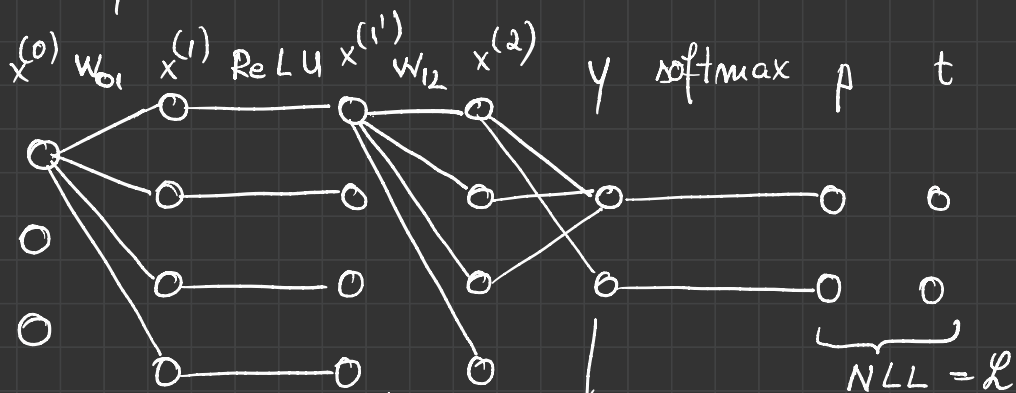
Diagram showing the derivation of the second term:

$$-\frac{\sum_{j'} e^{y_{j'}}}{e^{y_k}} \cdot \frac{e^{y_k} \cdot \sum_{j'} e^{y_{j'}} - (e^{y_k})^2}{\left(\sum_{j'} e^{y_{j'}}\right)^2} = \frac{e^{y_k}}{\sum_{j'} e^{y_{j'}}} - 1$$

Diagram showing the softmax function:

$$\frac{e^{y_k}}{\sum_{j'} e^{y_{j'}}} \rightarrow \text{softmax}$$

MLP final



$$\frac{\partial \mathcal{L}}{\partial y} = \frac{\partial \mathcal{L}}{\partial p} \cdot \frac{\partial p}{\partial y}$$

$$\frac{\partial \mathcal{L}}{\partial x^{(2)}} = \frac{\partial \mathcal{L}}{\partial y} \cdot \frac{\partial y}{\partial x^{(2)}}$$

$$\frac{\partial \mathcal{L}}{\partial w_{12}} = \frac{\partial \mathcal{L}}{\partial y} \cdot \frac{\partial y}{\partial w_{12}}$$